

Certificate of Analysis

PRODUCT DETAILS

Name

Product Name

Exosomes (RejuvCORE*)

Batch

Manufacturing Date

Expiry Date

Dosage Form

Clear-colourless solution

Storage Temperature

4 – 8°C

TEST	SPECIFICATION	TEST RESULTS	REMARKS
Infectious Diseases Screening	HBsAg; HCV Ab; HIV1/2 Ag/Ab; CMV Ab; RPR	Not Detected / Non-Reactive	QUALITY CONTROL APPROVED
Sterility	No cloudiness of the test media	Negative	QUALITY CONTROL APPROVED
Bacterial Endotoxin Test (LAL)	Endotoxin sensitivity 0.125 EU/mL	Negative	QUALITY CONTROL APPROVED
Mycoplasma	Mean of absorbance <0.05	Negative	QUALITY CONTROL APPROVED
Simple Western Blot	Positive Marker: Tetraspanin – CD9, CD63 Cytosolic – TSG101 Negative Marker: Cellular – Calnexin	Detected Trace marker detected	QUALITY CONTROL APPROVED
Nanoparticle Tracking Analysis (NTA)	Exosome particle size range 30nm – 200nm	10 x 10 ⁹ particles/mL	QUALITY CONTROL APPROVED

The cell product delivered by Malaysian Genomics Resource Centre Berhad is manufactured and tested for research purpose only. Product is processed in a facility operating under current Good Manufacturing Practice (cGMP) guidelines in accordance with PIC/S standards.

Storage recommendation:

- 1) Refrigerate the vials in 4-8°C at all times upon received
- 2) Keep away the vials from direct heat or sunlight
- 3) Avoid prolong exposure in room condition
- 4) It is recommended to place the vials in freezer for prolonged storage

*General Exosomes fortified with Curcumin

Authorised Signature:

Quality Control Department

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Supporting Information

STERILITY TEST ^{6, 8}

Aerobic microorganisms are incubated in Soybean Casein Digest Medium at 20°C - 25°C for 14 days.
Facultative microorganisms are incubated in Fluid Thioglycollate Medium at 30°C - 35°C for 14 days.

BACTERIAL ENDOTOXIN TEST (LIMULUS AMOEBOCYTE LYSATE, LAL) TEST ^{3,6}

Negative findings for Gram-negative bacteria in cell culture based on FDA Reference of Standard Endotoxin Guideline.

MYCOPLASMA TEST ^{4,6}

Negative findings for mycoplasma species *M. hyorhinis*, *M. arginini*, *M. fermentans*, *M. orale*, *M. Pirum*, *M. hominis*, *M. salivarium* and *A. laidlawii*.

SIMPLE WESTERN BLOT

Marker	Positive Marker		Negative Marker	
	Tetraspanin		Cytosolic	Cellular
	CD9	CD63	TSG101	Calnexin
Result	+	+	+	*

'+' indicates the presence of the marker in the cell.
'*' indicates the trace of the marker detected in the cell

NANOPARTICLE TRACKING ANALYSIS (NTA) ^{7,8}

Established method for the direct and real-time visualization and analysis of the nanoparticles size distribution and concentration in liquids.



REFERENCES

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4. US Pharmacopoeia <63> Mycoplasma Tests
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6. US Pharmacopoeia <71> Sterility Test
7. Neha Chopra, Ibrahim Dutt Arya, Namrata Jain, Poonam Yadav, Salma Wajid, Sarinder P. Singh, Sangeta Choudhary, 2019, Biophysical Characterization and Drug Delivery Potential of Exosomes from Human Wharton's Jelly-Derived Mesenchymal Stem Cells.
8. Dae Hyeon Hwang, Hyun-keun Kim, Joong Lee, Hyuck-Hoon Kwon, Gyeong-Hun Park, Seung-Hoong Yang, Jee Yoon Jung, Hyoung Choi, Jun Ho Lee, Suni Sung, Yong Weon Yi, Byung Seung Cho, 2020, Mesenchymal Stem/Stromal Cell-Derived Exosomes for Immunomodulatory Therapeutics and Skin Regeneration.

